Showing that the TKNN Integer is a Topological Invariant

As the senior thesis at NYU Shanghai

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1 Introduction

Topological insulators have been a focused physics problem these days, and the emergence of this field of study comes from the quantum Hall effect (QHE), which states that the conductance in the prescence of magnetic field is quantized. Haldane showed that even when the total magnetic flux is zero, there could be non-zero quantized conductance[1]. But the essence of the quantization was topological was not made clear yet. Until Berry put forward the notion of Berry connection form $\mathcal{A}$ as a function of the system(or the Hamiltonian) and Berry phase γ which in fact has geometric meanings: In adiabatic processes (that is to say when the system evolves slowly enough), one quantum state, say the n-th eigenstate stays in the n-th state rather than hopping to other states while gaining two phases $e^{i\theta_n}e^{\gamma_n}$[2]. The $e^{i\theta}$ phase is gained from the natural time-dependent Schrödinger's equation for a fixed state:

$$i\hbar\partial_t\psi_n = E_n\psi_n \Rightarrow \psi_n \mapsto \psi_n \exp\{(-i/\hbar)\int E_n(t)dt\} \equiv \psi_n e^{i\theta_n}, \tag{1}$$

while the geometric phase arises from the adiabatic process and takes form of

$$i\gamma_n := \int_C \mathcal{A}_n, \tag{2}$$

where C is some path in the parameter space and this Berry connection form is defined to be

$$\mathcal{A}_n := -i\left\langle d\psi_n(\mathbf{k}), \psi_n(\mathbf{k})\right\rangle; \quad \mathcal{A} := \sum_{j=1}^{n}\mathcal{A}_j. \tag{3}$$

Then TKNN (four people) showed that the quantization in the conductance arises from a formula, which is now known at the TKNN formula[3]:

$$\sigma = \frac{e^2}{2\pi h}\oint_C d\mathcal{A}, \tag{4}$$

where $\mathcal{A}$ is the Berry connection form and C is some closed curve in the parameter space.

The TKNN integer is defined to be

$$\frac{1}{2\pi}\oint_C d\mathcal{A}. \tag{5}$$

In 1983, Barry Simon pointed out that this integer is in fact related to the Chern class, since $\mathcal{A}$ gives a connection form and $d\mathcal{A}$ gives a curvature form and integrating over the curvature form yields a topological quantity[4]. But the detailed mathematics was not given there.

This paper aims to show that first, this integral does give rise to an integer as long as $H(k_x, k_y)$ is a hermitian operator; second, this integer is a topological quantity: That is to say, under continuous change of the function $H(k_x, k_y)$ (as long as the "gap" does not close), the value of this integer is not changed. In fact, this integer is the first Chern number of a hermitian bundle in a special case. In this paper, we will show with detailed calculation that this integer reflects the winding number (or known as the index) of a $U(n)$ phase.

The quantum Hall effect allows us to measure a more precise value of e^2/h, and has been a major way measuring the Planck's constant h. But this is not the most interesting part. Due to the topologcial nature of the quantized number, we developed a bulk-edge correspondance—for a material with non-zero Hall conductance, it is often insulating inside and conducting at the surface/edge. Such phenomenon is quite intriguing and worth digging into. Moreover, this fosters the emergence of Chern-Simons theory which is now used in topological quantum field theories.

We will adopt the Einstein summation notation throughout this paper. That it to say, we sum over repeated indices without writing the summation sign. The summation is over all possible values.

2 Vector Bundle

To deal with the problem, we must first get familiar with some knowledge of differential geometry and vector bundle.

Definition 1. *A vector bundle (of rank n) over a manifold M^m is a pair (E, π) where E is generally a set and π a projection map $E \to M$ such that*

$$
\begin{aligned}
&1. \quad \pi(E) = M; \\
&2. \quad \pi(p) = p, \quad \forall p \in M, \text{ is the fiber at } p; \\
&3. \quad \pi^{-1}(p) \text{ is an } n \text{ dimensional vector space}; \\
&4. \quad \forall p \in M, \exists U(p) \subset E \text{ diffeomorphic to } M \times \mathbb{F}^n, \\
&\qquad \text{the diffeomorphism } U(p) \to M \times \mathbb{F}^n \text{ is called a local trivialization at } p.
\end{aligned} \tag{6}
$$

One would at once realize the fourth condition is the same as the condition for a manifold, where we require the manifold is locally Euclidean and here we require the vector bundle the be locally trivial. For any $e \in E$, the fiber it lies on is some $\pi^{-1}(p)$, and this p is simply given by $\pi(e)$. So it makes sense to talk about the vector space e is on as well as the point from which e grows.

Just like in the case for a manifold, we may define a set of transition functions.

Definition 2. *For two trivializations ϕ_1 on $U_1(p)$ and ϕ_2 on $U_2(p)$, the change of coordinate function at p: $T_{12}(p) \in \mathbb{F}^n \to \mathbb{F}^n$ is defined to be $T_{12} := \phi_2 \phi_1^{-1}$.*

Since the trivializations are arbitrary: For example, multiplying the original trivialization by -1 still makes it a trivialization. However, the transition function is changed. So there are lots of choices of transition functions. A vector bundle is called "trivial" if the local trivialization can be made global. In other words, a trivial vector bundle is those diffeomorphic to $M \times \mathbb{F}^n$.

Proposition 1. *The change of coordinate functions of a trivial bundle E can be made identity maps.*

Proof. Let ϕ be the global trivialization then $\phi\phi^{-1}$ is always identity. $\square$

For those non-trivial vector bundles, however, we are not be able to find a global trivialization and we cannot make the change of coordinate functions identity.

Example 1. *The Möbius strip/bundle is a non-trivial bundle.*

Proof. To do this, we parametrize the Möbius strip: On a $\mathbb{S}^1$ grows strips that is tilting π gradually,

$$\begin{aligned} U &= \{(s,\theta) : -1 < s < 1, 0 < \theta < 2\pi\} \\ \phi_U^{-1}(s,\theta) &= ((1 - s\sin\tfrac{\theta}{2})\cos\theta, (1 - s\sin\tfrac{\theta}{2})\sin\theta, s\cos\tfrac{\theta}{2}) \\ V &= \{(s,\theta) : -1 < s < 1, -\pi < \theta < \pi\} \\ \phi_V^{-1}(s,\theta) &= t_U(s,\theta). \end{aligned} \tag{7}$$

When $s = 0$ they both reduce to $\mathbb{S}^1$ so they are indeed trivializations. The trivializations on patch 1: $\{-1 < s < 1, 0 < \theta < \pi\}$ are identical so the change of coordinate function is 1. The trivializations on patch 2 are represented by $\pi < \theta < 2\pi$ and $-\pi < \theta < 0$ correspondingly. We see that the first (s,θ) corresponds to the second $(-s, \theta - 2\pi)$. Hence the change of coordinate function is

$$T_{UV} = \phi_V\phi_U^{-1} : \quad (s,\theta) \mapsto (-s, \theta - 2\pi). \tag{8}$$

θ and $\theta - 2\pi$ correspond to the same point p on $\mathbb{S}^1$, and so $t_{UV}(p) : s \mapsto -s$. So $T_{UV}(p) = -1$. Since the change of coordinate function is 1 on patch 1 and -1 on patch 2, this function cannot be made identity. Hence the bundle is non-trivial. $\square$

Here we introduce another concept, which will make things easier since trivializations are in fact a little bit messier. But first, let us introduce the notion of sections. They are just analogous to the vectors fields.

Definition 3. *A section on E is a continuous mapping $\sigma : M \to E$ such that $\pi \circ \sigma = \mathrm{id}$.*

This is just saying that on each point, a vector is assigned to the fiber growing from that point. The image of the *zero section* is M itself.

Definition 4. *Let $U \subset M$ be an open neighbourhood. Choose n smooth sections $\{s_i\}_{i=1}^n$ of E on U such that they are linearly independent on U. Such a set of n sections is called a local frame field of E on U.*

We know that locally E looks like $U \times \mathbb{F}^n$, so that we can choose a set of basis for $\{p\} \times \mathbb{F}^n$ at every point $p \in U$. When such basis is continuous it becomes a section. Hence the local frame field is just a local basis for the fibers.

Definition 5. *Suppose we have two intersecting open neighbourhoods U, $V \subset M$, and on them we have two local frame fields $\{s_i^U\}$ and $\{s_i^V\}$. So on $U \cap V$ they are both well-defined, and they differ by a $U(n)$ transformation. This transformation is called the transition function t^{UV}. We hence have*

$$s_i^U = \sum_j t_{ij}^{UV} s_j^V. \tag{9}$$

We can see that the existence of frame field can be used to construct the trivialization maps, and one immediately notice that the trivialization constructed this way will induce a change of coordinate map, which is almost identical to our transition function. Therefore we can claim that the existence of a global frame field makes a bundle trivial and the non-existence makes a bundle non-trivial. To see these claims: Let $e \in E$, $p = \pi(e) \in U$ is the base of the fiber it is on. The local frame field on U enables us to express e in terms of $\{s_i\}$: $e = \alpha^i s_i$. And hence we may define the trivializations to be

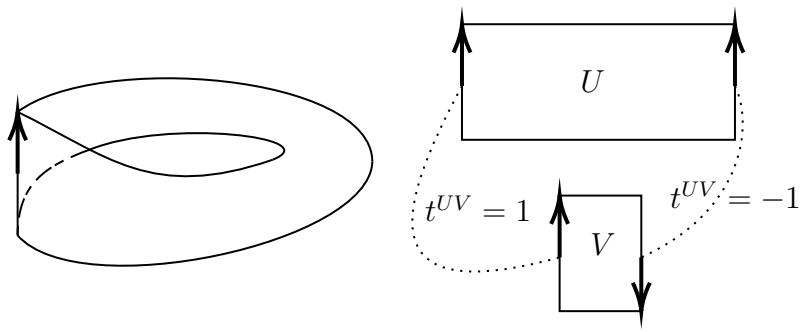
$$\phi_U : E|_U \to U \times \mathbb{F}^n, \quad e \mapsto (p, [\alpha_1, ..., \alpha_n]). \tag{10}$$

One immediately notice that $\phi_U s_i = (p, [0, ..., 1, ..., 0])$ where 1 is on the i-th entry. Hence to figure out the change of coordinate function (which is of course a matrix), we look at its actions on $[1, 0, ..., 0]$, etc. Now

$$\phi_V \phi_U^{-1}[0, ..., 1, ..., 0] = \phi_V s_i^U. \tag{11}$$

This is just to factor s_i^U into linear combinations of $\{s_i^V\}$, and is exactly what t^{UV} does. So

in such case, $T_{UV} = t^{UV}$. Therefore for a trivial bundle, we are able to find a global frame field of it and of course the transition function will be identity. To further understand this notion, we may look back into the case of the Möbius bundle:



Since on each patch, the local frame field consists of only one smooth section as the basis vector. They are either pointing "upward" or "downward" in the illustration above. WLOG, take the frame on U to be upward pointing. Suppose the frame on V is also pointing upward, since the "left" and "right" part of U and V are joined together with a flip, their left part coincide but their right part differ by a negative sign. Hence t^{UV} on the left intersection is 1 and is -1 on the right intersection. It is easy to see that such difference still exists if the frame on V is pointing downward. We can see that since the transition functions cannot be made uniformly 1, there cannot be a global frame field for the Möbius bundle and hence that bundle is non-trivial.

We can see that on the two intersections of U and V, the transition function takes discontinuous values: 1 and -1. If a point jumps from patch 1 to patch 2 and back to patch 1 forming a "loop", then the value of the transition function changes from 1 to -1 and back to 1. Topologically thinking, the value of the transition function winds around the origin for one turn. While for a trivial bundle, the value of the transition function stays constant 1, so it does not wind around the origin a bit—it winds for zero turn. So a non-zero winding number indicates a non-trivial bundle and a zero winding number indicates a trivial bundle. This winding number will be the essential *topological quantity* we are going to talk about. This idea will become more manifest later.

Now, we are at our last ingredient, which is an essential theorem classifying the topology of vector bundles. This theorem requires some technical constructions, and the thing we

need is one corollary in the end.

Definition 6. *Suppose we have $\pi : E \to M$ a vector bundle. For $f : N \to M$ we define the pullback bundle $f^{-1}(E)$ over N to be :*

$$\{(n, e) \in N \times E : \pi(e) = f(n)\}. \tag{12}$$

In other words, take any point $n \in N$, map this to $f(n) \in M$, on this point there is a fiber $\pi^{-1}(f(n))$, now grow this on N. This fiber bundle on N is $f^{-1}(N)$.

To think about this definition in a funnier way, we may think of f as a thief or a grabber, who grabs the fibers from M back.

Theorem 1. *Let N be a compact manifold, and we have f_0, f_1 homotopic maps from N to M, where we have a bundle E over X. Then $f_0^{-1}E \cong f_1^{-1}E$.*

This theorem says that if the thieves come from one class, they will steal the fibers in an isomorphic way. We will follow Hatcher's proof[5].

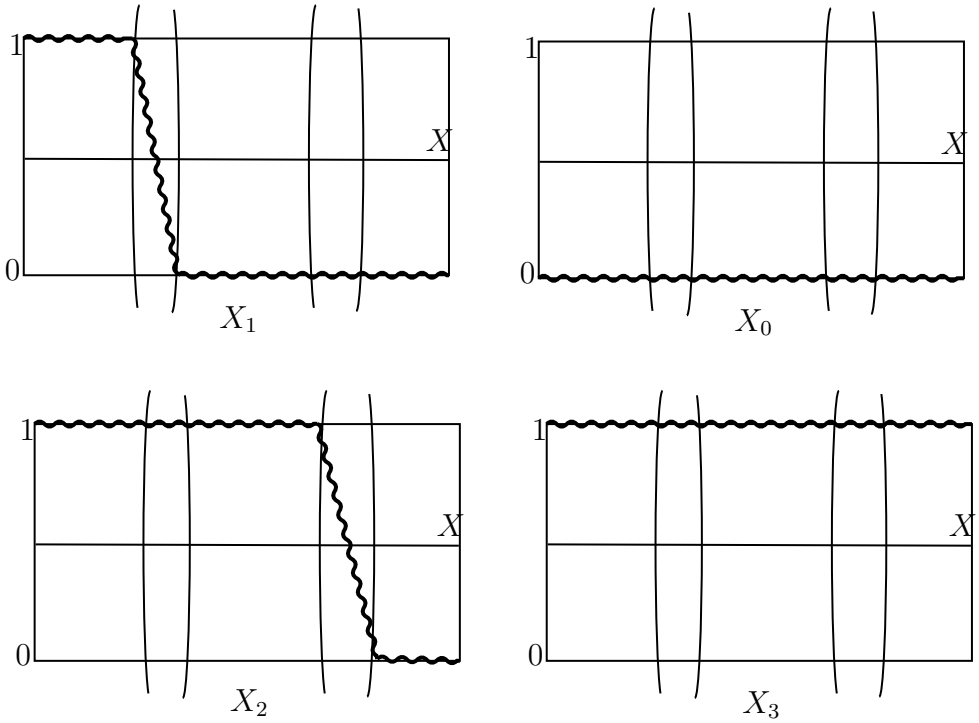
Lemma 1. *A rank-n vector bundle E over $X \times [a, b]$ is trivial if $X \times [a, c]$ and $X \times [c, b]$ are trivial.*

Proof. Let ϕ_1 be trivialize E_1 which is the restriction of E on $X \times [a, c]$ and let ϕ_2 trivialize E_2. We will construct a ϕ_2' that trivializes E_2 and agrees with ϕ_1 on $X \times \{c\}$. We notice that $\phi_1\phi_2^{-1}$ is a change of coordinate map defined on $X \times \{c\} \times \mathbb{F}^n \to X \times \{c\} \times \mathbb{F}^n$. And of course that map can be reduced to $\mathbb{F}^n \to \mathbb{F}^n$ and then extended to $T_{12} : X \times [c, b] \times \mathbb{F}^n \to X \times [c, b] \times \mathbb{F}^n$ without changing how it maps. Now, we define $\phi_2' := T_{12}\phi_2$, which is what we want. $\square$

Because the values of the trivializations at c only differ by a linear transformation, and multiplying a trivialization by that linear transformation still yields a trivialization.

Lemma 2. *There exists a set of trivializing maps for a bundle E over $X \times [0, 1]$ on $\{U_i \times [0, 1]\}_{i=1}^N$, where $U_i \subset X$.*

Proof. For each $x \in X$ we can find open neighborhoods $U_{x,1}, ..., U_{x,k}, ...$ in X and a partition $0 < t_0 < t_1 < ... < t_k = 1$ of $[0,1]$ such that the bundle is trivial over $U_{x,i} \times [t_{i-1}, t_i]$ since $[0,1]$ is compact. Now, by the previous lemma, the bundle is trivial over $U_{x,i} \times [0,1]$. $\square$



Lemma 3. *Let X be compact. The restrictions of a vector bundle E over $X \times [0,1]$ over $X \times \{0\}$ and $X \times \{1\}$ are isomorphic.*

Proof. By the previous lemma, the bundle can be chosen to be trivial on $\{U_i \times [0,1]\}$. Let $\{\psi_i\}$ be a partition of unity subordinate to $\{U_i\}$. Then define a sum of partition function $s_i(x)$ to be $\sum_{j \leq i} \psi_j(x)$. Now define $\{X_i\}$ to be $\{(x, s_i(x)) : x \in X\}$, which looks like the figure above. Of course, X_0 is just $X \times \{0\}$ and X_1 is $X \times \{1\}$.

We define $h_i : X_i \to X_{i+1}$ as $(x, s_i(x)) \mapsto (x, s_{i+1}(x))$. This map is a homeomorphism. Now we wish to define a homeomorphism H_i between $\pi^{-1}X_i$ and $\pi^{-1}X_{i+1}$, the bundles over X_i and X_{i+1}.

Since E is trivial over $U_i \times [0,1]$ for any i, any element in $U_i \times [0,1] \cap X_j$ can be written as $(x, s_k(x), v)_i$. (The subscript indicates that this is only locally true for index i.) We observe that if $i < j$, elements in $U_i \times [0,1] \cap X_j$ are all of $(x, 0, v)_i$, and if $i > j + 1$ those will be

$(x, 1, v)_i$. Therefore if $i < j$, $U_i \times [0,1] \cap X_j$ and $U_i \times [0,1] \cap X_{j+1}$ are exactly the same set. So the fiber on them are exactly the same. If $i > j+1$ the case is the same. Therefore we define our $H_i : \pi^{-1}X_i \to \pi^{-1}X_{i+1}$ function to be:

$$\begin{cases} \mathrm{id}, & \text{if } x \in U_j, \text{ where } j \leq i; \\ (x, s_j(x), v) \mapsto (x, s_{j+1}(x), v), & \text{if } x \in U_j, \text{ where } j = i+1; \\ \mathrm{id}, & \text{if } x \in U_j, \text{ where } j > i+1. \end{cases} \tag{13}$$

It is easy to check the continuity, and this function is a isomorphism. Since X is compact, the number of such X_i is finite, say N, and that $\pi^{-1}X_0$ is homeomorphic to $\pi^{-1}X_1$ through the map $H_0 H_1 \cdots H_N$. □

Now, the proof of the theorem is quite direct:

Proof. Consider f_t, the homotopy map between f_0 and f_1. Now we have a vector bundle $f^{-1}E$ over $N \times [0,1]$. And the restriction of $f^{-1}E$ on $N \times \{0\}$ is $f_0^{-1}E$, while the restriction of $f^{-1}E$ on $N \times \{1\}$ is $f_1^{-1}E$. They are isomorphic by the last lemma. □

Definition 7. $\mathrm{Vect}_k(M)$ *is the isomorphism classes of rank-k vector bundles over* M.

Corollary 1. *If* M *is homotopic to* N, $\mathrm{Vect}_k(M) \cong \mathrm{Vect}_k(N)$.

Proof. We have $f : M \to N$ and $g : N \to M$ such that $g \circ f$ is homotopic to id_M and $f \circ g$ is homotopic to id_N. Now for any element $E_M \in \mathrm{Vect}_k(M)$, $\{E_M\} = \{id_M^{-1} E_M\} \cong \{f^{-1}g^{-1}E_M\} \in \mathrm{Vect}_k(N)$. The isomorphism is $f^{-1}g^{-1}$. Its inverse is obviously $g^{-1}f^{-1}$, since $g^{-1}f^{-1}f^{-1}g^{-1}E_M \cong f^{-1}g^{-1}E_M \cong E_M$. □

3 TKNN Integer from a Detailed Perspective

In this section, $\mathbb{F} = \mathbb{C}$.

In the last section we already mentioned that our topological quantity is actually a winding number. Now we first clarify the notion of winding number:

Definition 8. *A closed curve f in $\mathbb{C}$ can be parametrized by $\mathbf{k} \in D$ as $f = r(\theta(\mathbf{k}))e^{i\theta(\mathbf{k})}$. The winding number around the origin is given by*

$$n(f) = \frac{1}{2\pi i} \oint_f \frac{dz}{z}. \tag{14}$$

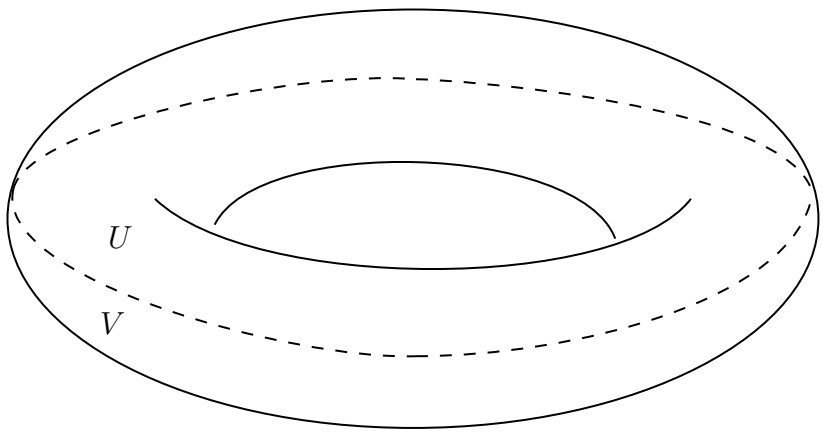
By residue theorem this number reflects the number of turns f winds around z_0. Of course, we calculate this integral using the parametrization

$$\frac{1}{2\pi i} \oint_f \frac{dz}{z} = \frac{1}{2\pi i} \int_D \frac{1}{f(\mathbf{k})} \nabla_{\mathbf{k}} f(\mathbf{k}) \cdot d\mathbf{k} = \frac{1}{2\pi i} \int_D \nabla_{\mathbf{k}} \log f \cdot d\mathbf{k}. \tag{15}$$

We denote the last expression as $W[f, D]$, while it is also our $n(f)$.

Now, we must clarify the background of the situation we will be dealing with. We have a manifold $M = \mathbb{R}/2\pi \times \mathbb{R}/2\pi$, a torus. It is parametrized by k_x,k_y and we write $\mathbf{k} := (k_x, k_y)$. On M we define an $\geq n$ dimensional hermitian operator $H(k_x, k_y)$, which is called the *Hamiltonian. This Hamiltonian H has exactly n eigenvectors corresponding the the eigenvalue* 0*, and all other eigenvalues are* 1. Now we define our rank-n fiber bundle E over M to be: $\forall p \in M, \pi^{-1}(p)$ is the eigenspace corresponding to eigenvalue 0. The topology we will be discussing is whether E is trivial. That is, whether there is a global frame field of E on M.

We cut the torus M in half into two patches U, V, each resembling a tubular neighbourhood of $\mathbb{S}^1$, as shown in the graph below. In the light of the following proposition, on U or V respectively we can have a *global* frame field. This fact is mentioned by Fu but not showed[6].



Proposition 2. *The complex bundle over $\mathbb{S}^1$ is trivial.*

By the homotopy theorem of vector bundles, isomorphism classes of rank-n vectors bundles over $\mathbb{S}^1$ and isomorphism classes of rank-n vector bundles over $\mathbb{S}^1 \times \mathbb{R}$ are the same. This is because $\mathbb{S}^1 \times \mathbb{R}$ and $\mathbb{S}^1$ are homotopy equivalent by contracting along $\mathbb{R}$. Therefore since the all complex vector bundle over $\mathbb{S}^1$ is trivial, we would have all complex vector bundle over $\mathbb{S}^1 \times \mathbb{R}$ be trivial.

Proof. A $\mathbb{C}$-vector bundle over $\mathbb{S}^1$ can be constructed by the following gluing the two endpoints of $[0,1]$ together:

$$[0,1] \times \mathbb{C}^n / \{(0,v) \sim (1,Tv)\}, \tag{16}$$

where $T \in \mathrm{GL}(n,\mathbb{C})$. Let E_T denote this vector bundle.

Lemma 4. *If T_0 and T_1 are homotopic by a family of invertible linear transformations on $\mathbb{C}^n$, then $E_{T_0} \cong E_{T_1}$.*

Proof. Let $T_t \in \mathrm{GL}(n,\mathbb{C})$ be the homotopy. The first vector bundle is $[0,1]/\{(0,v) \sim (1,Tv)\}$. Now I claim that E_{T_0} and E_{T_1} are isomorphic through the map

$$E_{T_0} \to E_{T_1}, \quad [t,v]_{T_0} \mapsto [t, T_t(v)]_{T_1}. \tag{17}$$

This map is clearly an isomorphism. At the gluing point, howvever, we must check the

well-definedness and bijectiveness:

$$[0, v]_{T_0} \mapsto [0, T_0 v]_{T_1}; \quad [1, T_0(v)]_{T_0} \mapsto [1, T_1 T_0 v]_{T_1} \sim [0, T_0 v]_{T_1}. \tag{18}$$

Hence the map is well-defined.

□

In the light of the above lemma, we now show that any such vector bundle is isomorphic to the trivial bundle, E_I. To show this we only need to show that there is a homotopy between any $T \in \mathrm{GL}(n, \mathbb{C})$ and I. In other words, we need to show that $\mathrm{GL}(n, \mathbb{C})$ is simply-connected: Let $T \in \mathrm{GL}(n, \mathbb{C})$. By Schur decomposition, T can be written in the following matrix form

$$T = Q \begin{bmatrix} \lambda_1 & & & A \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} Q^{-1}. \tag{19}$$

The eigenvalues are non-zero. Then our first path would be to multiply A part in the matrix by $(1-t)$ for this matrix is connected to a diagonal one. Then the second path would be to replace each eigenvalue $\lambda_j = r_j e^{\theta_j}$ with $[(1-t)r_j + t]e^{(1-t)\theta_j}$, so that this diagonal matrix is connected to the identity. The essence is that any nonzero *complex* number is connected to 1 avoiding 0 in $\mathbb{C}$. □

This proposition is understandable to me thanks to the help of Professor C. K. Fok.

Lemma 5 (Jacobi's Formula)**.**

$$\frac{d \det(A)}{d\alpha} = \det(A) \operatorname{tr}\left(A^{-1} \frac{dA}{d\alpha}\right). \tag{20}$$

So we have

$$\partial_\alpha \log(\det A) = \sum_{i,j} (A^{-1})_{ij} \partial_\alpha A_{ji}. \tag{21}$$

Proof. We know that $\det(I+\epsilon H) = 1 + \epsilon \mathrm{tr} H + \mathcal{O}(\epsilon^2)$, so

$$\begin{aligned}
\det(A(\alpha+\epsilon)) &= \det(A + \frac{dA}{d\alpha}\epsilon + \mathcal{O}_{\epsilon^2}) \\
&= \det(A)\det\{A^{-1}[A + \epsilon(\frac{dA}{d\alpha} + \mathcal{O}_\epsilon)]\} \\
&= \det(A)\det[I + \epsilon A^{-1}(\frac{dA}{d\alpha} + \mathcal{O}_\epsilon)] \\
&= \det(A)[1 + \epsilon \mathrm{tr}(A^{-1}\frac{dA}{d\alpha} + \mathcal{O}_\epsilon)] + \mathcal{O}_{\epsilon^2} \\
&= \det(A) + \epsilon \det(A)\mathrm{tr}(A^{-1}\frac{dA}{d\alpha}) + \mathcal{O}_{\epsilon^2}.
\end{aligned} \tag{22}$$

Therefore

$$\frac{d\det(A)}{d\alpha} = \det(A)\,\mathrm{tr}\left(A^{-1}\frac{dA}{d\alpha}\right). \tag{23}$$

Furthermore

$$\partial_\alpha \log(\det A) = \frac{1}{\cancel{\det A}}\cancel{\det(A)}\,\mathrm{tr}\left(A^{-1}\frac{dA}{d\alpha}\right) = \sum_{i,j}(A^{-1})_{ij}\partial_\alpha A_{ji}. \tag{24}$$

□

Now, finally, we are here to reveal the topological significance of the TKNN integer. We will adopt a physicists' notations on vectors. The local frame basis vectors will be normalized and denoted by $|i,U\rangle$ substituting s_i^U. The dual vectors will be denoted by $\langle i,U|$ substituting $\left\langle\cdot, s_i^U\right\rangle$. From the knowledge of vector bundles, we know that

$$\langle n,V|m,U\rangle = t_{mn}^{UV}. \tag{25}$$

Here $t^{AB} \in U(n)$ is the transition function, we have

$$(t^{UV})^\dagger t^{UV} = I \quad \Longleftrightarrow \quad (t_{ij}^{UV})^* t_{ik}^{UV} = \delta_{jk}. \tag{26}$$

Be careful that t_{mn}^{UV} is just a number and commutes with anything. The *Berry connection*

A is defined as

$$iA^U = \sum_m \langle m, U| \nabla_{\mathbf{k}} |m, U\rangle . \tag{27}$$

Then we have

$$\begin{aligned} iA^U &= \sum_m \langle m, U| \nabla_{\mathbf{k}} \sum_n t^{UV}_{mn} |n, V\rangle \\ &= \sum_m \sum_n \langle m, U| (\nabla_{\mathbf{k}} t^{UV}_{mn}) |n, V\rangle + t^{UV}_{mn} \langle m, U| \nabla_{\mathbf{k}} |n, V\rangle \\ &= \sum_{m,n} (t^{UV}_{mn})^* \nabla_{\mathbf{k}} t^{UV}_{mn} + \sum_{m,n} t^{UV}_{mn} \langle m, U| \nabla_{\mathbf{k}} |n, V\rangle . \end{aligned} \tag{28}$$

Here the second term is given by

$$\begin{aligned} \sum_{m,n} t^{UV}_{mn} \langle m, U| \nabla_{\mathbf{k}} |n, V\rangle &= \sum_{m,n} \sum_l t^{UV}_{mn} \left(t^{UV}_{ml}\right)^* \langle l, V| \nabla_{\mathbf{k}} |n, V\rangle \\ &= \sum_{n,l} \delta_{n,l} \langle l, V| \nabla_{\mathbf{k}} |n, V\rangle \\ &= iA^V . \end{aligned} \tag{29}$$

Also by Jacobi's formula,

$$\sum_{m,n} (t^{UV}_{mn})^* \nabla_{\mathbf{k}} t^{UV}_{mn} = \mathrm{tr}[(t^{UV})^\dagger \nabla_{\mathbf{k}} t^{UV}] = \nabla_{\mathbf{k}} \log(\det t^{UV}). \tag{30}$$

Therefore we arrive at

$$A^U - A^V = -i\nabla_{\mathbf{k}} \log(\det t^{UV}). \tag{31}$$

One would notice immediately that this is the integrand in the expression of the winding number defined before. Now we may take a closed curve $C \subset U \cap V$. We would have

$$\int_C (A^U - A^V) \cdot dl = \int_C -i\nabla_{\mathbf{k}} \log(\det t^{UV}) = 2\pi W[\det t^{UV}, C]. \tag{32}$$

If $C = \partial U = -\partial V$ with $U \cup V = M$. Then

$$\int_C A^U \cdot dl = \int_U \nabla \times A^U \cdot d\mathbf{k}; \quad \int_C A^V \cdot dl = -\int_V \nabla \times A^V \cdot d\mathbf{k}. \tag{33}$$

We verify that $\nabla \times A^U = \nabla \times A^V$, and we define that vector to be $\mathcal{F}$, the *Berry curvature*. Then

$$\begin{aligned}\int_M \mathcal{F} \cdot d\mathbf{k} &= \int_U \mathcal{F} \cdot d\mathbf{k} + \int_V \mathcal{F} \cdot d\mathbf{k} \\ &= \int_{\partial U} (A^U - A^V) \cdot dl \\ &= 2\pi W[\det t^{UV}, \partial U].\end{aligned} \tag{34}$$

Therefore

$$\frac{1}{2\pi} \int_M \mathcal{F} \cdot d\mathbf{k} = W[\det t^{UV}, \partial U] = -W[\det t^{VU}, \partial V]. \tag{35}$$

If this vector bundle is trivial. This is to say that we can have a global frame field. So the frame on patch U is the same as the one on patch V in their intersection. This would mean that $t^{UV} = I$, and $W[\det t^{UV}, \partial U] = W[1, \partial U] = 0$. Hence a trivial vector bundle corresponds to a zero winding number, and a non-trivial vector bundle would necessarily correspond to a non-zero winding number. This number essentially reflects the topology of the vector bundle—how twisted it is.

4 Chern Class

The above calculations are detailed and "makes our hands dirty," but it showes the twistedness in a clearer way. Of course, to show that the TKNN integer is a topological quantity and truly is an integer, we may adopt some more advanced mathematics in differential geometry—the Chern class. This notion is not a coincidence. Since we are integrating over a 2 dimensional manifold without boundary, the result is the same if the integrands differ by any exact form. This is exactly the cohomology group $H^2(M)$, and the first Chern class is an element in it.

We define two states are parallel if they can be adiabatically transported to each other, then the parallel transport yields

$$\psi_n(t) = \psi_n(0) \exp \int \mathcal{A}_n, \tag{36}$$

where the Berry connection form $\mathcal{A}_n = A_n \cdot dl$ for an energy eigenstate $|n\rangle$ is

$$\mathcal{A}_n := -i \langle n|d|n\rangle , \tag{37}$$

and so the parallel transports reads

$$|n\rangle \mapsto |n\rangle \exp\{\int -i \langle n|d|n\rangle\}. \tag{38}$$

An infinitesimal transport would lead to

$$|n\rangle \mapsto |n\rangle (1 - i \langle n|d|n\rangle). \tag{39}$$

So the connection matrix is $\omega^i{}_j = \delta^i{}_j \mathcal{A}_i$ (not summed). The curvature form is simply

$$\Omega^i{}_j = d\omega^i{}_j + \sum_k \omega^i{}_k \wedge \omega^k{}_j = \delta^i{}_j d\mathcal{A}_i + \sum_k \delta^i{}_k \mathcal{A}_i \wedge \delta^k{}_j \mathcal{A}_j = \delta^i{}_j d\mathcal{A}_i. \tag{40}$$

And since the first Chern class is

$$c_1 = \frac{i}{2\pi}\mathrm{tr}(\Omega) = \frac{i}{2\pi}\sum_n (-i) d\left\langle n|d|n\right\rangle\} = \frac{1}{2\pi}\sum_n d\left\langle n|d|n\right\rangle . \tag{41}$$

We immediately know that

$$\mathcal{F}\cdot d\mathbf{k} = \nabla\times A\cdot d\mathbf{k} = dA\wedge dl = d\mathcal{A} = d\sum_n \left\langle n|d|n\right\rangle = \sum_n d\left\langle n|d|n\right\rangle . \tag{42}$$

So the integral of

$$\frac{1}{2\pi}\oint_{[-\pi,\pi]^2}\mathcal{F}\cdot d\mathbf{k} = \oint_{[-\pi,\pi]^2} c_1 \in \mathbb{Z}. \tag{43}$$

is the first Chern number.

Here the adiabatic theorem provides us with a diagonal connection matrix, so that Ω is simply $d\omega$. Such connection is called a canonical connection induced by a Hermitian norm in the paper by Chern and Bott[7]. Hence we can now see that our topological quantity is the first Chern number, and that from the previous section we can now know that the first Chern number reflects the twistednees of a Hermitian bundle.

5 Applications to Physics

First of all, these applications are in the field of condensed matter physics, and we introduce the concepts of insulators and conductors. They are characterized by the band theory in Physics. In quantum mechanics there is a famous relation called the de Brogile relation, which states that the momentum p_x is $\hbar k_x = \partial_{k_x} H(k_x, k_y)$ for a free electron. Here in a more general system, we have

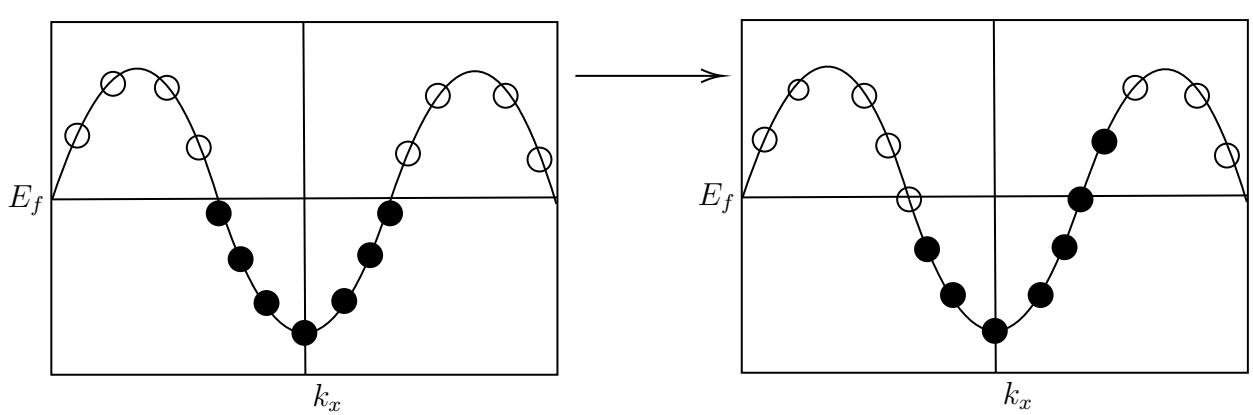
$$\langle v_x \rangle = \frac{1}{m} \langle p_x \rangle = \left\langle \frac{1}{\hbar} \frac{\partial H}{\partial \mathbf{k}} \right\rangle . \tag{44}$$

So for an eigenstate $\psi_n(\mathbf{k})$, it contributes to a current

$$\mathbf{j} = 2e \langle v_n \rangle = \frac{2e}{\hbar} \left\langle \frac{\partial E_n}{\partial \mathbf{k}} \mathbb{1}_{E_n(\mathbf{k}) \leq E_f} \right\rangle . \tag{45}$$

The existence of the last term $\mathbb{1}_{E_n(\mathbf{k}) \leq E_f}$ indicates that the wavefunction is partially filled for some $\mathbf{k}$. Here E_f is the Fermi energy, a constant. Our Hamiltonian is always an even function in $\mathbf{k}$, and so this average is zero. However, when under an electric field $\mathbf{E} = E\hat{x}$, Drude's model tells us that the $\mathbf{k}$ are shifted by an extra momentum $\Delta\mathbf{k} = eE\Delta t/\hbar$. So now the indicator function becomes $\mathbb{1}_{E_n(\mathbf{k}-\Delta\mathbf{k}) \leq E_f}$ and the current becomes

$$\mathbf{j} = \frac{2e}{\hbar} \left\langle \frac{\partial E_n}{\partial \mathbf{k}} \mathbb{1}_{E_n(\mathbf{k}-\Delta\mathbf{k}) \leq E_f} \right\rangle . \tag{46}$$



Therefore a partially filled eigenstate which is also called band, since it looks like a band,

contributes to the current. This therefore leads to the definition of insulators and conductors:

Definition 9. *A Hamiltonian H describes a conductor if it has a partially filled band (eigenstate). Since all bands under E_f are all occupied, only the top band could be partially filled. Therefore if the top band crosses other bands, it has to be a conductor.*

Definition 10. *A Hamiltonian H describes an insulator if all its occupied bands are fully filled. Since all bands under E_f are all occupied, it equals to saying all occupied bands are under E_f. Therefore insulators have a gap between the n-th and $(n+1)$-th band:*

$$\inf_{\mathbf{k}} E_{n+1} - \sup_{\mathbf{k}} E_n > 0. \tag{47}$$

First we must make it clear what it implies by being a topological invariant—this number is left unchanged under continuous change of what it depends on. In our case the invariant depends on $H(k_x, k_y)$.

Four people TKNN showed that the Hall conductivity in a material is given by

$$\sigma = \frac{e^2}{2\pi h} \int_{[-\pi,\pi]^2} d\mathcal{A}, \tag{48}$$

which now we know can be expressed by

$$\sigma = n\frac{e^2}{h} \tag{49}$$

where n is the topological invariant we just computed. This quantity being topological indicates that if the Hamiltonian can be continuously changed to

$$0\sum_{i\leq n} |i\rangle\langle i| + 1\sum_{i>n} |i\rangle\langle i| \tag{50}$$

without closing a gap in the highest energy. The last sentence is needed because if the n-th band touches the $n+1$-th band, we will not be able to define the n-subspace of eigenstates at that energy crossing point, for there will be $n+1$ states.

Since any gap closing comes with two bands (else we cannot cross), label the bands to

be i and $i+1$. If $i+1 \leq n$, following the above arguments we see that the Chern number is well-defined. If $i > n$ the Chern number is well-defined since they do not take part in the integration. If $i = n$, however, the Chern number is not well-defined according to the above arguments. Therefore when the gap closes for the n-th band i.e. the top band, the topological quantity may not be preserved. *Therefore for system with a valid Chern number, there is always a finite gap between the n-th and $(n+1)$-th band.* Such systems are characterized as topologically insulating.

Definition 11. *A Hamiltonian H describes a topological insulator if there is a* direct *gap between the n-th and $(n+1)$-th band:*

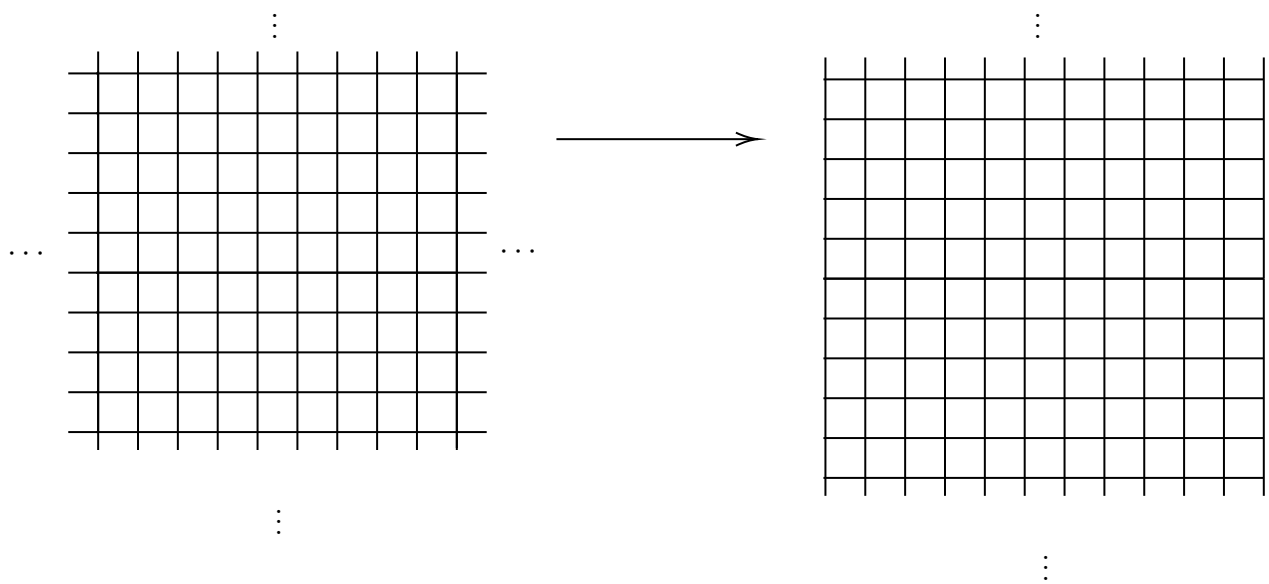
$$\inf_{\mathbf{k}}(E_{n+1} - E_n) > 0. \tag{51}$$

Of course, this system could have been conducting. But usually (and for more interesting phenomenon) it is insulating. If the Chern number is 0, we call it a trivial topological insulator; if the Chern number is non-zero, we call it a non-trivial topological insulator.

Lemma 6. *Suppose H_t is a one-parameter family of the Hamiltonians and H_0 is topologically insulating and is of zero Chern number and H_1 is topologically insulating and is of non-zero Chern number. Then there must exist some $0 < \tau < 1$ such that H_τ is conducting i.e. has the top band crossing.*

Now, this notion leads to a famous bulk-edge correspondance.

First, we have to know what is an edge state. Since $H(k_x, k_y)$ is periodic in k_x and k_y, it in fact describes a material which extends periodically and infinitely. We call it the bulk state. Sometimes however we kill the periodicity in, say, the k_x direction.



Now this new system is $\tilde{H}(k_y, N)$ where N is the number of stacks in the x direction. It is illustrated by the graph on the right. The Chern number is defined for the system illustrated on the left.

Theorem 2. *If an insulating system $H(k_x, k_y)$ has a non-zero Chern-number, $\tilde{H}(k_y, N)$ will be conducting for N large enough, and the conducting state is localized at the edge, which is called an edge state.*

Proof. First consider the layer in the vertical middle. Since it is so far away from the boundary we may not distinguish it from the bulk. If we add x periodicity to it, it will have the same Chern number as the bulk does—non-zero. Next we consider the layer closer to the edge. Since such process is very guadual, the system will have only changed a bit so the Chern number is still protected. This process proceeds until we reaches the edge, since outside the material is the vaccum which of course has a zero Chern number. The Chern number has changed and so there must have been a top band crossing at the place where the Chern number changed. That is the edge of the material. □

Therefore all non-trivial topological insulator has a gap closing in the top band, which is localized at the edge. Other bands that are away from the edge cannot touch higher bands.

Such key feature is a feature of non-trivial topological insulators. Such materials have been tested to exist in 1978 by Wakabayashi and Kawaji[8].

In fact, apart from this TKNN integer, physicists have developed more topological quantities like the $\mathbb{Z}_2$ integer which classifies non-trivial topological insulators in time-reversal invariant systems (there are $2n$ occupied states considering spin). Such integer is quite similar to our case. Rather than measuring the existence of a global continuous basis, it measures the existence of a global continuous basis under the condition that the basis consists of

$$|1, \mathbf{k}\rangle, ..., |n, \mathbf{k}\rangle, \mathcal{T}\,|1, -\mathbf{k}\rangle, ..., \mathcal{T}\,|n, -\mathbf{k}\rangle \tag{52}$$

where $\mathcal{T}$ is the time-reveral operator. To characterize this we may wish to divide the torus into two parts and see whether in the two parts we have global continuous n eigenstates. This is discussed with details by Liang Fu. He also studied another $\mathbb{Z}_2$ integer which classifies non-trivial topological insulators in C_4 rotationally invariant systems in a similar pattern. We can see that many modern physically topological quantities are characterizations of different vector bundles.

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